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Tutorial group: TX

Matriculation number:

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17 September 2019

MH2500 Test 1

40 minutes

QUESTION 1.**(10 marks)**Suppose that the cumulative distribution function (CDF) of X is given by

$$F(x) = \begin{cases} 0, & x < 1; \\ 1/3, & 1 \leq x < 2; \\ 5/6, & 2 \leq x < 4; \\ 1, & x \geq 4. \end{cases}$$

- (a) (2 marks) Find $\Pr\{1 < X < 2\}$ and $\Pr\{2 \leq X < 3\}$. (No partial credits will be given for each probability.)
- (b) (8 marks) Compare $\log_2 \mathbb{E}(X)$ with $\mathbb{E}(\log_2 X)$. Which is larger? State your reason.

[Solution:]

Observe that X only takes discrete values 1, 2, 4, and $\Pr\{X = 1\} = 1/3$, $\Pr\{X = 2\} = 1/2$ and $\Pr\{X = 4\} = 1/6$. Thus

(a) $\Pr\{1 < X < 2\} = 0$, $\Pr\{2 \leq X < 3\} = 1/2$ and $\Pr\{X \geq 2\} = 2/3$.

(b) $\mathbb{E}(X) = 1 \cdot (1/3) + 2 \cdot (1/2) + 4 \cdot (1/6) = 2$ and so $\log_2 \mathbb{E}(X) = 1$.

$$\mathbb{E}(\log_2 X) = 0 \cdot (1/3) + 1 \cdot (1/2) + 2 \cdot (1/6) = 5/6.$$

$$\text{Hence } \log_2 \mathbb{E}(X) > \mathbb{E}(\log_2 X).$$

QUESTION 2.**(8 marks)**

Let events A and B be independent, with $\Pr(A) = p$ and $\Pr(B) = 0.3$. Let C denote the event that at least one of the events A and B occurs and D denote the event that exactly one of the events A and B occurs.

- (a) (2 marks) Find $\Pr(C)$. Express your answer in terms of p .
- (b) (3 marks) Find $\Pr(D)$. Express your answer in terms of p .
- (c) (3 marks) Can C and D be independent? If this is possible, find all possible values of p such that C and D are independent; otherwise state your reason why C and D can never be independent.

[Solution:]

- (a) $C = A \cup B$, so

$$\Pr(C) = \Pr(A) + \Pr(B) - \Pr(A \cap B) = \Pr(A) + \Pr(B) - \Pr(A) \Pr(B) = 0.3 + 0.7p.$$

- (b) $D = C \setminus (A \cap B)$, so

$$\Pr(D) = \Pr(C) - \Pr(A) \Pr(B) = 0.3 + 0.4p.$$

- (c)

$$\begin{aligned} C \text{ and } D \text{ are independent} &\Leftrightarrow \Pr(C \cap D) = \Pr(C) \Pr(D) \\ &\Leftrightarrow \Pr(D) = \Pr(C) \Pr(D) \\ &\Leftrightarrow \Pr(C) = 1, \end{aligned}$$

where the last line follows from the fact that $\Pr(D) > 0$. Hence when $p = 1$, C and D are independent.

QUESTION 3.**(11 marks)**

Customers arrive at a shop randomly with a rate of 4 per hour, modelled by a Poisson process. In a one-hour period, calculate the conditional probability that no customers arrive in the second half hour given that

- (a) (4 marks) exactly 4 customers arrive in the first half hour;
- (b) (7 marks) exactly 4 customers arrive over the whole one-hour period.

[Solution:]

- (a) Let X be the number of customers who arrive in the second half hour, then $X \sim \text{Poisson}(2)$. By independence with the customers arriving in the first half hour,

$$\Pr\{X = 0\} = e^{-2}.$$

- (b) Let Y be the number of customers who arrive in the first half hour and N be the number of customers who arrive over the one-hour period, then $Y \sim \text{Poisson}(2)$ and $N \sim \text{Poisson}(4)$.

$$\Pr\{X = 0|N = 4\} = \frac{\Pr\{X = 0 \text{ and } N = 4\}}{\Pr\{N = 4\}} \quad (1)$$

$$= \frac{\Pr\{X = 0\} \Pr\{Y = 4\}}{\Pr\{N = 4\}} \quad (2)$$

$$= \frac{e^{-2} \cdot \frac{2^4}{4!} e^{-2}}{\frac{4^4}{4!} e^{-4}} \quad (3)$$

$$= \frac{1}{16}. \quad (4)$$